



Sharing Session by Operation Hope Foundation and HRRP-Nepal

Low Cost Housing Solutions for Vulnerable HHs in Himalayan Regions of Nepal

28th August 2020



PRESENTERS

- Mr. **Robert Kee**, Chairman & Founder, Operation Hope Foundation.
- Mr. **Wesley Goh**, Project Manager, Hope Trust Nepal.

FACILITATOR

- Mr. **Minar Thapa Magar**, HRRP, National Coordinator-General Coordination.

ORGANIZERS

- Operation Hope Foundation.
- Housing Recovery and Reconstruction Platform (HRRP) - Nepal.

*Link to complete webinar recording [here](#).

LOW COST HOUSING SOLUTIONS FOR VULNERABLE HHs IN HIMALAYAN REGIONS OF NEPAL

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This sharing session held to share the experiences by Operation Hope Foundation (OHF) in constructing more than 150 affordable Rice Bag House (RBH) in Dhading, Bhojpur and Kavre Districts. The event provided a platform to share and interact on the implementation, affordability, technical feasibility and lessons on constructing low cost housing in Himalayan regions. The focus was to hear the experiences and to explore other ideas of low-cost housing solutions to provide the vulnerable beneficiaries with less financial capacity to reconstruct their houses with the government grant. This program was organized by the Housing Recovery and Reconstruction Platform (HRRP) – Nepal and Operation Hope Foundation, and was facilitated by Minar Thapa Magar, HRRP, National Coordinator-General Coordination.

The sharing session began with the introduction and presentation of how “Low-Cost Housing Solutions for Vulnerable” households, done by Operation Hope Foundation in the Himalayan regions of Nepal. A comprehensive view of the technical, social and cultural considerations, and affordability of RBH were discussed in the session.

The sharing session was facilitated by HRRP and had two presenters:

Presenters:

- Mr. Robert Kee, Chairman & Founder, Operation Hope Foundation.
- Mr. Wesley Goh, Project Manager, Hope Trust Nepal.

Facilitator:

- Mr. Minar Thapa Magar, HRRP, National Coordinator-General Coordination.

Organizers:

- Operation Hope Foundation.
- Housing Recovery and Reconstruction Platform (HRRP) - Nepal.

KEY POINTS HIGHLIGHTED

- Rice Bag House (RBH) as an affordable building solution for the Himalayan areas.
- OHF experience in constructing RBH for earthquake affected vulnerable households in Nepal and integrating livelihood support within the project lifecycle.
- Lessons learnt (Technical and Engineering aspect)
- Extending support to low-cost housing solutions for supporting Vulnerable beneficiaries left behind.

AGENDA

- Rice Bag House (RBH)
- OHF Experience with RBH -Dhuseni(Below 1 Lakh) -Bhojpur(DUDBC Design)
- Other Application -Community Hall in Okladhungha
- Supporting the Vulnerable left behind
- Technical Advice

DISCUSSION

After the 2015 Earthquake, the need for temporary housing in an emergency situation was very high. The use of Rice Bag House (RBH) started for the victims in Dhading district. The technique proved to be affordable (below 1 Lakh) and durable. Two years after the earthquake, RBH was included in the DUDBC catalogue under “Design Catalog Volume 2” as Earthbag Masonry. With experience gained from previous field work, OHF proceeded to build more RBH in Bhojpur within 3 Lakhs grant from government and it also built a 1,200 sq. ft. community hall in Okladhungha using the same technology. Presenter Mr. Robert Kee highlighted the community involvement in building low cost houses using RBH technique. Some key points of the presentation, discussion and its feasibility in context of Nepal, are as follows:

Building Solution for Himalayan area

- Cooler in Summer, suitable for small/terrace land as there is no need for big footers (foundation), no need of carrying heavy bricks up in the mountain but with just empty rice bags for construction, storage loft available and no requirement of skilled masons (for flat roof house). This is a new skill for the community to learn and support each other to rebuild.

Experience in Dhading

- Stone in mud mortar (traditional) houses were destroyed during the earthquake. Sites were inaccessible by vehicle. Villagers recycled their old windows, doors,

zinc roof and collected stones and gravel from the river. They used the traditional method of mud and cow dung mixture for plastering. A 200 sq.ft. house cost less than NPR 30,000 with the labour support from the community.

- Villagers were innovative and they learned the skill to build using this technique. Some built a porch, a pitch roof, with chicken coop on the 2nd storey, and painted the structure in bright colours.

Experience in Bhojpur

- RBH was included in the DUDBC catalogue under “Design Catalog Volume 2” as Earthbag Masonry approved by the government where beneficiaries are eligible for the 3 lakhs grant. As villagers are vulnerable (illiterate and poor), most have not even taken the 1st tranche. Replicating the community project led by the local leaders, each villager played their part to help each other.
- Cost of the 200 sq. ft. DUDBC house was only 2.4 lakhs (including tools, transport of bulk material and food for builders). With the grant of 3 lakhs, villagers can find carpenters and build a better-quality roof.
- RBH appeared like a normal brick house after plastering and paint. The houses of big lofts with a pitch roof design.

Community Hall

- In Okhaldhunga, a community hall was damaged by the earthquake. The community requested to build a new one which would be used for community meetings, training and health services post serving 500 households. The proposed design and plan with local stakeholders took 1 month to complete the 1,200 sq.ft. hall.
- The Rice Bag House was very suitable in the 35 km of rough terrain, as transportation cost for normal housing was very high.

Mr. Wesley Goh shared the second presentation, which was to share and discuss the technical details, manuals, guidelines and feasibility of RBH. He highlighted some key points as given below:

Technical Sharing

Debunking the myth on the durability of RBH:

- RBH foundations are deep and strong. They are packed with 1-meter big stones and 0.5 meter gravel inside geotextile. Perforated pipes are embedded for water drainage to prevent any soil erosion.
- Rice bags are load bearing walls. Each bag weighs 50 kg providing a strong

downward force.

- Barbed wire is laid twice around the walls at each layer and rebars inserted every 2 feet to tackle the lateral force (horizontal movement) of the house and make it earthquake resistant.
- Rice bags are made of Polypropylene which is non-biodegradable but breaks down under UV exposure. Without exposure to UV rays (covered in mud/plaster) polypropylene bags are reported to remain intact as long as 500 years.
- Rice bags are also commonly used as emergency flood walls due to its strong tensile strength.

Technical Advice

- Wire mesh wrapped around rice bags needs to be tight and taut for easy application of plaster. Cement to sand ratio must be 1:4 with the sand sieved for a smooth finish. Mud can be used to fill in the gaps between rice bags to reduce the use of cement.
- Hook/ tie the roof structure / frame to the rice bag. As each rice bag weighs around 50 kg, tying it to 3 layers at each point is equivalent to holding the roof down with nearly 1 tonne of weight.
- Gaps in-between the roof and walls reduce the insulation capability of RBH. Using the flexibility of rice bags various shapes and sizes can be created to close any gaps in-between.
- Free instructional video and step-by-step manual are available online at <http://www.ohf.org.sg/rbh-downloads>

OPEN FLOOR DISCUSSION AND SOME KEY VIEWPOINTS

After the sharing of presentation on low cost housing solution from Operation Hope Foundation, an open discussion session was carried out where several queries were put forward. "Lot of photos from Dhuseni has mud or cow dung plaster. While the presentation slide shows the use of right mixture of plaster for sustainability. The DUDBC catalogue mentioned only the use of cement sand plaster but not of mud or cow dung plaster due to the limited cost on providing socio-technical assistance. As per OHF, it was primarily because of the budget constraint and the use of cow dung and mud is the traditional method of plastering in the community and thus, a quick result was achieved. The chicken wire framing of the rice bag wall is much better and stronger but with the increment of cost of construction. This technology is suitable for earthquake regions and for flood-prone areas. Earth bags are used as flood barriers, gravels are filled in bags instead of mud/sand in flood prone areas.

Build Up Nepal (Pratik Singh): CSEB (Compressed Stabilized Earth Blocks) may be an alternative to RBH and promote entrepreneurship among the community. The transportation of materials at Himalayan region is difficult but also an earthquake resilient house needs to be built under minimum budget. CSEB with a requirement of sand, mud and cement (10%) incurs the cost of a 2-roomed house, approximately costing 3.2 to 3.5 million NPR.

NSET (Aamod Mani Dixit): The need in Nepal is varied. It may be used in schools, temples, houses, hospitals, etc. All of these technologies are of great value and incomparable in terms of their functionality and benefit. These are particularly good for the vulnerable beneficiaries. The question is of durability and stability. How long is the rice bag or rock bag durable? This technology can even be adopted not just by vulnerable households but even by the other socio-economic classes. Many people have no knowledge of it and only their life can teach them experience of it. Lastly, it must undergo laboratory tests to identify its strength and durability. He also raised questions regarding the stability of bags and its durability, and has also suggested that if it passes all the tests then this technology should be applied to other beneficiaries as well.

Central Level Project Implementation Unit (CLPIU-Building) -Jhapper Singh Vishokarma- Deputy Project Director: It is useful for vulnerable beneficiaries. DUDBC has also published an NRA catalogue to support beneficiaries who want to construct their houses using this technique. But the seismic analysis is not yet carried out. A query was put on the durability of such rice bag houses built, as experienced at Sindhupalchowk district, where the houses were attacked by rats and insects disturbing the building. The rice bags were damaged by these intruders therefore he raised the question on how can we get rid of these problems? He also asked for the comfort that the beneficiary can get using this technique. How environmentally friendly is it?

SAND/RICE-BAG HOUSE BUILDING RELATED ISSUES

- One of the positive aspects of this type of building is the temperature controlling mechanism in both the summer and winter season.
- Is there a comparison between rammed earth and sandbag technology?
- How durable is such a building and how to repair and maintain this type building was also one of the issues raised?

ANNEX A: QUESTIONS AND ANSWERS (Q&A)

Q: Why mud/cow dung was used instead of cement for plastering the walls?

A:

Mud & cow dung are used purely to lower down the cost and the main purpose of the project was to use the locally available materials which could slump down the cost of construction rather than cement, and it was easy to use mixture of mud and cow dung with the locally available masons.

Q: Is it suitable for flood prone areas?

A:

Yes it is suitable for the flood prone areas too but instead of sand in the bag we can use gravel and stone rubble available locally.

Q: Any seismic analysis report available?

A:

A professor from National University of Singapore and Nanyang Technology University have both assessed that the design is earthquake resistant. They have done a miniature RBH with a shaker and it is durable (available on our website). Nepal did a seismic test and then the design was inserted in the DUDBC catalogue. Another NGO built an Earthbag school before 2015 and it is still intact after the earthquake. which provides some evidence that it is resistant to seismic loads.

Q: What is the frequency of repair and maintenance?

A:

Ensure that the rice bags are covered by plaster. Intact plaster won't expose the rice bags to UV rays, and it will last very long. For flood prone areas, ensure that the water level will not be higher than the gravel bags, so it allows the water to seep through and drain out properly. if all these criteria are maintained then it remains longer and there is no need of frequent repair and maintenance.

Q: Any cost estimation?

A:

For the DUDBC approved design, cost estimation just for materials is 2.3 lakh NPRs (including door and windows but not labour.) Detailed breakdown can be found on the website:

<http://www.ohf.org.sg/rbh-downloads>

Q: Although skill masons are not needed, some skills are still required. Do OHF plans to do training camps?

A:

Yes. We welcome training camps where 20 people will participate and will complete one house during the training. It's important to build the first house together with the villagers so they know how to do it properly. They may proceed to form groups and build their own house after the training. As per Robert Kee, they had also prepared an hour video which is in Nepali from where people can learn about this technology.

Q (Subash Sharma): Does the roof with a CGI sheet make an RBH house thermally comfortable?

A:

Robert: *A laminated roofing sheet was used to prevent heat and fire from transferring into the room which can lower the temperature inside during summer and keeps warmer inside during the winter season, as well as for sound too.*

Q (Sneha Malani): Many himalayan pockets use some form of rammed earth for construction. Earthbag masonry is a variation of the same. How would you compare rammed earth with earthbag masonry in terms of its scalable use in remote areas of Nepal? Have you tried plastering in cement stabilized earth instead of cement plaster all together?

A:

Robert: *They have no experience of rammed earth use so far. The advantage of earthbags is you don't require formwork, which is required to build rammed earth walls, and can incur more costs.*

Q (Sufi M. Faiz): What challenges do you face during the construction? How did you convince the community about the new technology? As there is perception that brick and stones are stronger for construction.

A:

Robert: *it was really challenging to convince the community with their cultural acceptance on the use of new technology. With a completion of 2 houses, OHF had to wait for 3 months and finally it took 6 months to convince 55 HHs to build the house.*

Q: What is the range of life span of low-cost housing that we are promoting in Nepal? What is the minimum to maximum cost of two-story buildings in Kathmandu Valley?

A:

The life span of such a type of housing is about 500 years with proper technology applied as per the guidelines but it may degrade if exposed to UV rays. Therefore, precision must be applied with plastering.

Q: What could be the possible low-cost housing solution for flood affected areas of Nepal?

A:

For flood affected areas sand should be replaced by gravel which will not be wasted away even when the water shives in a bag from the plaster.

Q: I am working as technical facilitator engineer in a rural housing project in Gorkha district, so want to know about that typology for support for the vulnerable in my project areas. CR technology in this regard.

A:

You can take support of the guideline at this website. <http://www.ohf.org.sg/rbh-downloads>

Q: The scope of Compressed Stabilised Earth Bricks and entrepreneurship approach for vulnerable households in the area.

A:

CSEB may be an alternative to rice bags and promote entrepreneurship among the community. The transportation of materials in Himalayan regions is difficult, but also an earthquake resilient house needs to be built under minimum budget. CBSE with a requirement of sand, mud and cement (10%) where the cost of a 2-room house is 3.2 to 3.5 million NPR.

Q: What is the comfort level and designs available? What is the requirement of land for a typical family of five?

A:

As gaps in between the roof and walls reduce the insulation capability of RBH, it is very comfortable for living (7 degree difference in temperature). The design of RBH is available in the DUDBC catalog under "[Design Catalog Volume 2 as Earthbag Masonry](#)". Beneficiaries can build RBH as per the needs of the family following NRA guidelines.

Q: It will be good to know the tentative Budget Figure of the Low-Cost House.

A:

It is as low as NPR 30,000 meeting the NRA approved design to receive 3 lakhs government grant.

Q: Has the vulnerable HH list prepared by NRA really covered the most vulnerable ones in the community?

A:

Yes. The NRA published the list of 18505 vulnerable beneficiaries covering all earthquake-affected districts, which contains the most vulnerable households and families that need to be addressed and provided with low cost housing solutions.

Q: How the rock bag house is constructed and what about the seismic performance of those houses?

A:

Professor from National University of Singapore and Nanyang Technology University have both assessed that the design is earthquake resistant. Have done a miniature RBH put on shaker and it's durable (available on our website). Nepali did a seismic test and thus design was inserted in the DUDBC catalogue. Another NGO built an Earthbag school before 2015 and it was still intact after the earthquake. which provides some evidence that it is resistant to seismic loads.

Q: How to scale up low cost housing solutions as an available option for all those who cannot afford to rebuild otherwise? (reach all those left behind so far)

A:

The beneficiary and community must be well aware and have enough knowledge on RBH technology through which low cost housing solutions can scale up.

Q: What is the minimum cost of housing that has been constructed in the Himalayan region of Nepal after the 2015 earthquake?

A:

2.3 to 2.4 lakhs NPR that include salvage doors and windows with some tools and CGI sheets excluding labour cost.

Q: Can we make it more secure and have fire cooking inside the house?

A:

Yes.

Q: How durable is RBH construction? What is the frequency of maintenance and repair?

A:

The life span of such a type of housing is about 500 years with proper technology applied as per the guidelines but it may degrade if exposed to UV rays. Therefore, precision must be applied with plastering. If the RBH is properly plastered there is very less maintenance of repair needed in future.

Q: What are the primary areas of RBH houses that require repair in future?

A:

Wall is the primary area if not treated well in case of plastered well it acts as a normal building.

CLOSING

CLPIU Building-Shyam Kishor Singh (Project Director) concluded the sharing session with thanks to HRRP Team and OHF for sharing their experience on RBH technology that would be very helpful for the vulnerable beneficiaries. He added that the design of RBH is available in DUDBC catalogue under “Design Catalog volume 2” as Earthbag Masonry and it is very helpful for vulnerable people especially in remote areas where economic support and durability are their main concern.

He further mentioned that the low-cost housing was his major concern in the reconstruction to achieve the goal of “Leave no one behind” and this sharing session has helped us all to think further on it. He concluded that about 6 million houses need to be built out of which 50% HHs have been built. The remainder can be quickly achieved using such techniques.

ANNEX B: PARTICIPATION DETAILS

Country	Registered	Attended
Bangladesh	1	0
Canada	1	0
Greece	1	0
Malaysia	2	2
Nepal	104	91
Philippines	4	2
Singapore	1	1
Thailand	1	0
United Kingdom	3	1



Geographic distribution of the webinar participants

SN	Name	Affiliation	Country
1	Amanda Duff	DFID	United Kingdom
2	Ambika Amatya	Catholic Relief Services	Nepal
3	Amod Mani Dixit	National Society for Earthquake Technology-Nepal (NSET)	Nepal
4	Babita Sharma	Transparency International Nepal	Nepal
5	Badri Pyakurel	Housing Recovery and Reconstruction Platform (HRRP)	Nepal
6	Balkrishna Gaire	Nepal National Disabled Association	Nepal
7	Bansha Lal Tamang	Indrawati Rural Municipality	Nepal
8	Beth Simons	CARE International UK	United Kingdom
9	Bhubaneswari Parajuli	HRRP	Nepal
10	Bibek Chapagain	Royal Consultants P. LTD.	Nepal
11	Bibek Shrestha	CRS	Nepal
12	Bimal Kathayat	Lawyer	Nepal
13	Bishwas Thapa	CRS	Nepal
14	Buddhi Bal Magarati (Mijar)	Mott MacDonald Limited Company Nepal Branch	Nepal
15	Danu Darai	HRRP Nepal	Nepal
16	Deepak Aryal	CRS	Nepal

SN	Name	Affiliation	Country
17	Deepak Dhungel	AIDS Healthcare Foundation (AHF) Nepal	Nepal
18	Deepak Saud	HRRP	Nepal
19	Dikendra Raj Kandel	COPPADES	Nepal
20	Dinesh Adhikari	Global IME Bank Ltd.	Nepal
21	Dr. Sunit Adhikari	SAPPROS Nepal	Nepal
22	Durga Prasad Dhakal	ChildFund Japan	Nepal
23	Dwarika Shrestha	Department of Buildings	Nepal
24	Fran Lloyd	Nepal School Projects	Canada
25	Francis Baraoidan	Habitat for Humanity Int'l Asia Pacific	Philippines
26	Ganesh Paudel	Freelancer	Nepal
27	Giselle Goh	TBC Foundation S B	Malaysia
28	Goh Chin Leng, Wesley	OHF	Nepal
29	Govinda Bhattarai	OVN	Nepal
30	Guheswari Tuladhar	Urban Environment Management Society (UEMS)	Nepal
31	Gun Bir Newar	Foundation for International Development/Relief (FIDR)	Nepal
32	Hari Chandra Satyal	UNOPS	Nepal
33	Hari Devkota	Team of volunteers	Nepal
34	James Tamang	Naitik Bikash Sewa Kendra(Bhaisenpati, Lalitpur)	Nepal
35	Janak Sharma	Caritas Nepal	Nepal
36	Jhapper Singh Vishokarma	NRA, CLPIU (Building)	Nepal
37	Jim Kendall	Habitat for Humanity	Philippines
38	Kailash Khadka	UNOPS	Nepal
39	Kamal Bahadur Shrestha	NHRP/UNDP	Nepal
40	Kamran Akbar	World Bank	Nepal
41	Kirti Chandra Joshi	HRRP	Nepal
42	Kong Yee Wong	Volunteer worker	Malaysia
43	Kriti Rajkarnikar	Geometrix Pvt. Ltd.	Nepal
44	Kshitiz Paudel	Abhiyaan Engineering Consultants And Construction Private Limited	Nepal

SN	Name	Affiliation	Country
45	Kshitiz Shrestha	Building Design Authority Pvt. Ltd.	Nepal
46	Kushal Chaudhari	United Nations Resident Coordinator's Office (UNRCO)	Nepal
47	Lajana Manandhar	Lumanti Support Group for Shelter	Nepal
48	Lena Michaels	The Asia Foundation	Nepal
49	Mahesh Chitrakar	MoUD, Department of Urban Development and Building Construction (former)	Nepal
50	Mahesh Paudel	NRA DLPIU LALITPUR	Nepal
51	Makhan Maharjan	Urban Environment Management Society (UEMS)	Nepal
52	Mehul Savla	Catholic Relief Services	Bangladesh
53	Merina Bajracharya	UNOPS	Nepal
54	Monika Shrestha	CRS/HRRP	Nepal
55	Narayan Lamichhane	DCA	Nepal
56	Nishan Shrestha	ChildFund Japan	Nepal
57	Nishan Shrestha	ChildFund Japan	Nepal
58	Nurlan Moldosherip	Habitat For Humanity International Asia Pacific	Philippines
59	Phaindra Pandey	CRS	Nepal
60	Pia Jensen	Habitat for Humanity	Philippines
61	Prabhat Kumar Thakur	NRA DLPIU (BUILDING), NUWAKOT	Nepal
62	Prabin Gautam	CRS/HRRP	Nepal
63	Pradip Sagar Tiwari	Opportunity Village Nepal	Nepal
64	Pradip Sharma	HRRP	Nepal
65	Prakash Khadka	Jcycn	Nepal
66	Pranav Dahal	NSET	Nepal
67	Prashant Gautam	Kathmandu University	Nepal
68	Pratik Singh Parmar	Build up Nepal	Nepal
69	Rabindra Das	Transform Nepal	Nepal
70	Radhe Shyam Rai	Organizational Development Support	Nepal
71	Raj Babu Shrestha	UDEN	Nepal
72	Rajendra Chiluwal	NRA, Nuwakot	Nepal

SN	Name	Affiliation	Country
73	Rajesh Sunuwar	CRS	Nepal
74	Rajesh Sunuwar	CRS	Nepal
75	Raju Pradhan	Caritas Germany	Nepal
76	Ranjan Dhungel	NSET	Nepal
77	Raul Gautam	National Reconstruction Authority	Nepal
78	Rijan Gajurel	HRRP-CRS	Nepal
79	Rita Thakuri	NSET	Nepal
80	Robert Kee	Operation Hope Foundation	Singapore
81	Robert Subedi	Church	Nepal
82	Roshan Raj Giri	NRA,CLPIU(BUILDING)	Nepal
83	Rupesh Kumar Yadawa	EHRP/JICA	Nepal
84	Ruplal Aidi	Housing Recovery and Reconstruction Platform (HRRP) implemented by Catholic Relief Services (CRS)	Nepal
85	Sagun Dongol	Swastik Builders Pvt. Ltd.	Nepal
86	Sajina Sherpa	Catholic Relief Services	Nepal
87	Salena Sangachhe	ASF Nepal	Nepal
88	Sampad	DFID Field Office	Nepal
89	Samriddhi Sharma	UNRCO	Nepal
90	Saroj Dahal	DFID Field Office	Nepal
91	Seb Kaminski	Arup	United Kingdom
92	Seki hirano	CRS	Greece
93	Shankar Pokhrel	NRA	Nepal
94	SHANTA RANA	UNOPS	Nepal
95	Shiba Regmi	National Association of Rural Municipalities in Nepal (NARMIN)	Nepal
96	Shiva Hari Devkota	Habitat for Humanity International Nepal	Nepal
97	Shuva Sharma	Scott Wilson Nepal (SW Nepal Pvt Ltd)	Nepal
98	Smriti Kayastha	Bellissimo Design International	Nepal
99	Sona Bade	National Reconstruction Authority	Nepal
100	Sudeep Acharya	DLPIU KTM	Nepal
101	Sudin Pradhan	Free lance	Nepal

SN	Name	Affiliation	Country
102	Suman Chandra Gurung	Birat Nepal Medical Trust (BNMT)	Nepal
103	Suman Khadka	Organization Development Support	Nepal
104	Sunil Khanal	Habitat for Humanity International Nepal	Nepal
105	Suraj Shrestha	CARE Nepal	Nepal
106	Surendra Bahadur Bam	Care Nepal	Nepal
107	Suresh Kattel	Catholic Relief Services	Nepal
108	Suresh Nakarmi	Habitat for Humanity Nepal	Nepal
109	Surya Man Gurung	company	Nepal
110	Sushil Ojha	Shinawatra University	Thailand
111	Suvash Gautam	Emergency Housing Reconstruction Project	Nepal
112	Tulaja Maleku	Dlpiu kathmandu chandragiri	Nepal
113	Udaya Bikram Thapa	World Vision International Nepal	Nepal
114	Udaya Bikram Thapa	World Vision International Nepal	Nepal
115	Udhab Raj Subedi	Catholic Relief Service Nepal	Nepal
116	Uttam Paudel	HRRP	Nepal
117	Yadav Mainali	Save the Children	Nepal
118	Yubaraj Bhandari	HRRP Nepal	Nepal

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